

temperature is 288.2 K

A1 [3]

4 (a) (i) 1.0

B1 [1]

(ii) 40 Hz

B1 [1]

(b) (i) speed = $2\pi fa$
= $2\pi \times 40 \times 42 \times 10^{-3}$
= 10.6 m s^{-1}

C1

A1 [2]

(ii) acceleration = $4\pi^2 f^2 a$
= $(80\pi)^2 \times 42 \times 10^{-3}$
= 2650 m s^{-2}

C1

A1 [2]

(c) (i) S marked correctly (on 'horizontal line through centre of wheel')

B1

(ii) A marked correctly (on 'vertical line' through centre of wheel)

B1 [2]

Page 2	Mark Scheme	Syllabus
	GCE A/AS Level – May/June 2006	9702

5 (a) (i) force per unit positive charge (ratio idea essential)